Frayer models

SECTION A / VOCABULARY Name _____

Find the word on the screen. Talk it through with your partner, then write or sketch in its four boxes.

Distance vs. displacement

Meaning in my words How would you explain the difference between distance and displacement?	**How can you tell?** Which quantity can be zero after a walk that covers six metres?
Example or sketch Walk 4 m east and 1 m west. What distance and displacement would you record?	**Non-example and why** Someone reports 5 m east as the displacement. Which part of the return trip did they miss?

Net force

Meaning in my words How would you explain net force to a friend?	**How can you tell?** Two people push one cart in opposite directions. Why do the arrow directions matter?
Example or sketch One person pushes 6 N right and another 4 N left. What single arrow replaces both?	**Non-example and why** A student adds those pushes to get 10 N right. Which push did they count the wrong way?

Frayer models

SECTION A / VOCABULARY Name _____

Find the word on the screen. Talk it through with your partner, then write or sketch in its four boxes.

Coordinate sign

Meaning in my words How would you explain coordinate sign to a friend?	**How can you tell?** If right is positive, what sign goes on a 3 N force left?
Example or sketch How would you label a leftward velocity of 2 m/s when right is positive?	**Non-example and why** Does a negative velocity by itself mean the cart is slowing down?

Interaction pair

Meaning in my words How would you explain interaction pair to a friend?	**How can you tell?** Earth pulls an apple down. What is the partner force, and which object feels it?
Example or sketch How would you label the pair when a hand pushes a cart?	**Non-example and why** Are the table's push and Earth's pull on one book a third-law pair?

Doodle the idea

SECTION B / DRAW WITH YOUR TEACHER Name _____

Draw a force sum that includes a leftward force. Circle the first place where losing the sign would change your answer.

First wrong step

Repair

My marks mean

Stop the faulty graphic

19

SECTION E FINAL MEETING / TEAM TRANSFER

Name _____ Partner _____

Our goal: Use all forces to calculate acceleration.

CHOOSE ONE JOB TOGETHER

SOFTWARE TESTER ☐

Mark the first wrong calculator step. Send the developer the corrected force sum and expected acceleration.

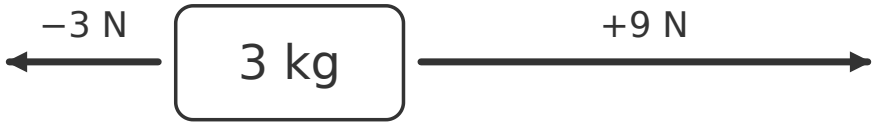
FACT-CHECK PRODUCER ☐

Stop or clear the graphic. Give the anchor a corrected number that viewers can check against the force arrows.

Point to a clue. Keep your first marks; add the update using the color key.

A calculator overlay for a **3 kg** tracker shows **+9 N** and **–3 N**. Its output is "**9 ÷ 3 = 3 m/s².**" It is about to go live.

DISPLAY: 9 ÷ 3 = 3 m/s²



01 OUR FIRST CALL

Can this calculation go live? Give the corrected result and explain.

02 OUR REVISED CALL

Wait for the reveal. Keep your first call.

New clue: _____

Did the developer fix the error? Send a follow-up that names what still needs changing.

Switch speakers. Did the calculation include both forces?

☐ GREEN	☐ RED	☐ YELLOW	☐ BLUE	☐ PURPLE	☐ ORANGE	☐ PENCIL
speeding up	slowing down	steady / balanced	position / displacement	velocity	forces	other marks